Notes: Bown and Crowley (AER, 2013)

Self-Enforcing Trade Agreements: Evidence from Time-Varying Trade Policy

Contents

1 Big Picture

One-sentence summary. This paper takes the Bagwell and Staiger (1990) repeated-game theory of self-enforcing trade agreements to the data by asking whether temporary U.S. antidumping and safeguard tariff increases are more likely after import surges, especially in products with larger terms-of-trade incentives and less volatile import growth.

Research question. Why do governments sometimes raise tariffs even while they are members of cooperative trade agreements such as the WTO? Are temporary tariffs merely political-economy protection, or can they also be understood as equilibrium adjustments that help sustain self-enforcing cooperation after trade-volume shocks?

Core idea. In the Bagwell-Staiger model, a cooperative low-tariff agreement is sustained by the threat of reverting forever to a noncooperative high-tariff Nash equilibrium. When an import surge occurs, the importing country has a stronger one-period temptation to defect because a tariff can generate a larger terms-of-trade gain. To preserve cooperation, the most cooperative sustainable tariff may need to rise temporarily.

Empirical setting. The authors study U.S. temporary import tariff increases under antidumping and safeguard laws from 1997 to 2006. The panel is country i by industry k by year t: 49 U.S. trading partners, 283 NAICS manufacturing industries, and annual observations.

Main empirical predictions.

- **Import-surge prediction:** a tariff increase is more likely when recent bilateral imports rise.
- **Market-power prediction:** conditional on an import surge, a tariff increase is more likely when import demand and foreign export supply are more inelastic.
- **Unusual-surge prediction:** conditional on a given import surge, a tariff increase is more likely when the sector's historical import growth is less volatile, because the surge is more unusual.

Main finding. The data strongly support these predictions. In the baseline antidumping specification, a one-standard-deviation increase in import growth raises the likelihood of a tariff by about 35 percent; a one-standard-deviation increase in the inverse elasticity measure raises it by about 88 percent; and a one-standard-deviation increase in import-growth volatility lowers it by about 76 percent.

2 Incremental Contribution of the Paper

2.1 Contribution relative to trade-agreement theory

Before this paper, Bagwell and Staiger (1990) provided a powerful theory of managed trade and self-enforcing trade agreements, but the central intertemporal predictions had not been directly tested. Bown and Crowley make the theory empirically operational. The key move is to interpret antidumping and safeguard tariffs not simply as political protection, but as observable time-varying tariff instruments that may respond to the same economic forces emphasized by the theory.

2.2 Contribution relative to empirical trade-policy literature

Earlier empirical work on trade policy often emphasized cross-sectional tariff levels or political-economy determinants such as industry concentration, employment, or injury indicators. This paper differs in three ways.

1. It focuses on **changes in tariffs over time**, not only levels of tariffs across products.
2. It links tariff changes to **import-volume shocks**, which are central to the repeated-game theory.
3. It directly incorporates **terms-of-trade incentives** through import demand and export supply elasticities.

2.3 Contribution relative to political-economy explanations

The paper does not reject political economy. Instead, it shows that the economic variables implied by self-enforcing agreement theory remain quantitatively important after adding standard political-economy controls. This matters because it suggests that U.S. temporary tariffs are not only explained by domestic lobbying or injury criteria; they also line up with the logic of sustaining cooperation under the threat of defection.

3 Theory: Self-Enforcing Trade Agreements

3.1 Repeated-game logic

The paper builds on a two-country repeated trade-policy game. In a one-shot game, each large country has an incentive to impose inefficiently high tariffs or export taxes because it can improve its terms of trade. In a repeated game, lower cooperative trade policies can be sustained if the one-period gain from defection is smaller than the discounted value of future cooperation.

Let $\Omega^*(\cdot)$ denote the importing country's one-period gain from defecting from the cooperative agreement, and let $\omega^*(\cdot)$ denote the discounted future gain from maintaining cooperation. The no-defection condition is

$$\Omega^*(k, k^*, \tau^c(k, k^*), \tau^*_D(k, k^*, \tau^c(\cdot))) \leq \omega^*(\tau^c(k, k^*), \tau^*_c(k, k^*)). \tag{1}$$

Interpretation. A cooperative tariff is sustainable only if the importing country does not want to deviate today. If an import shock raises the immediate benefit of defection, then the cooperative tariff may have to rise to reduce the temptation to defect.

3.2 The one-period gain from defection

Start from a cooperative free-trade equilibrium, $\tau^*_c = 0$ and $\tau^c = 0$. If the importing country deviates to its best-response tariff τ^*_D while the exporter stays cooperative, the one-period gain from defection is

$$\Omega^*(k,k^*,0,\tau_D^*) = \underbrace{[P^f - P(k,k^*,0,\tau_D^*)]M(k^*,P^*(k,k^*,0,\tau_D^*))}_{\text{terms-of-trade gain}} - \underbrace{\int_{P^f}^{P^*(k,k^*,0,\tau_D^*)} [M(k^*,P^*) - M(k^*,P^*(k,k^*,0,\tau_D^*))}_{\text{domestic efficiency loss}} \tag{2}$$

Economic message. A tariff is attractive because it lowers the exporter price and shifts surplus toward the importing country. But it is costly because it distorts the domestic consumption price and reduces trade volume. The incentive to defect is stronger when the terms-of-trade gain is large relative to the efficiency loss.

3.3 Import surge and elasticity prediction

Bagwell and Staiger show that the defecting incentive rises after a positive import-volume shock if the efficiency cost is sufficiently small:

$$\frac{d\Omega^*(\cdot)}{dk^*} > 0 \quad \Longleftrightarrow \quad \frac{\partial M(k^*,P^f)}{\partial k^*}\left[\frac{P^f}{\eta_x^f + \eta_m^f}\right] > \int_{P^f}^{P^*(k,k^*,0,\tau_D^*)} \frac{\partial M(k^*,P^*)}{\partial k^*} dP^*. \tag{3}$$

The left-hand side captures the terms-of-trade benefit from an import expansion. The key term is

$$\frac{1}{\eta_x^f + \eta_m^f}, \tag{4}$$

where η_x^f is the free-trade export supply elasticity and η_m^f is the free-trade import demand elasticity. If both elasticities are high, the importing country has little market power, and tariff manipulation is less attractive. If the elasticities are low, the country can move prices more and the incentive to raise tariffs is stronger.

3.4 Gains from cooperation and the role of import variance

In the special linear case,

$$M(k^*,P^*) = k^* - aP^*, \qquad X(k,P) = k + aP, \tag{5}$$

free-trade volume is $V^f = (k+k^*)/2$. The expected discounted gains from cooperation can be written as

$$\omega^*(\tau^c(V^f)) = \frac{\delta}{1-\delta}\left\{\frac{5}{12a}\left(\sigma_{V^f}^2 + [EV^f]^2\right) - \frac{a}{4}\left(\sigma_{\tau^c}^2 + [E\tau^c(V^f)]^2\right)\right\}. \tag{6}$$

The most cooperative tariff rule is

$$\tau_c^*(V^f,\omega^*) = \tau^c(V^f,\omega^*) = \begin{cases} 0, & V^f \in [0,\bar{V}^f], \\ \dfrac{1}{2a}(V^f - \bar{V}^f), & V^f \geq \bar{V}^f, \end{cases} \tag{7}$$

where $\bar{V}^f = \sqrt{6a\omega^*}$ is the trade-volume cutoff below which the most cooperative tariff is free trade.

Key comparative static. For two sectors with the same import surge, the tariff increase should be larger in the sector with lower import variance. A surge in a usually stable sector is more unusual and therefore implies a stronger temptation to defect.

4 Empirical Design

4.1 Estimating equation

The authors aggregate the theoretical predictions into the following empirical model:

$$y_{ikt} = \beta_0 + \beta_1 M_{ikt} + \beta_2 \left(\frac{1}{\eta_{xk} + \eta_{mk}} \right) + \beta_3 \left(M_{ikt} \times \frac{1}{\eta_{xk} + \eta_{mk}} \right) + \beta_4 \sigma_{ik}^m + \varepsilon_{ikt}. \tag{8}$$

Here:

- y_{ikt} is an indicator for whether the United States imposes an antidumping or safeguard tariff against country i in industry k after an investigation initiated in year t.
- M_{ikt} is the recent change in imports of industry k from country i.
- $1/(\eta_{xk} + \eta_{mk})$ measures the strength of the terms-of-trade incentive.
- $M_{ikt} \times 1/(\eta_{xk} + \eta_{mk})$ allows import surges to matter more when the terms-of-trade motive is stronger.
- σ_{ik}^m is the standard deviation of import growth, capturing whether an import surge is unusual.

The model is estimated using probit and logit specifications for tariff incidence and Tobit specifications for tariff size.

4.2 Why antidumping and safeguards are useful empirical objects

Antidumping and safeguard tariffs are not ordinary bound tariffs. They are temporary trade barriers that WTO members can impose under certain conditions. This makes them a natural empirical counterpart to time-varying cooperative tariffs: they are legally permitted ways for a government to raise trade restrictiveness without fully abandoning the trade agreement.

4.3 Identification logic and limitations

The paper is not a randomized experiment. Its identification comes from testing precise comparative statics across time, countries, and industries. The strongest part of the design is that the theory predicts a joint pattern: tariff increases should rise with import growth, rise with the inverse elasticity measure, and fall with import-growth volatility. Political-economy variables are added to test whether these theoretical variables survive standard alternative explanations.

A limitation is that import growth may be correlated with unobserved political pressure or injury. The authors address this by adding industry concentration, employment, value added over shipments, inventories over shipments, and sector indicators for steel and chemicals. The core results remain.

5 Data

The panel covers 49 U.S. trading partners, 283 NAICS manufacturing industries, and the years 1997-2006. Because explanatory variables require lags, the estimated policy data run from 1999 to 2006.

Main data sources.

- Temporary Trade Barriers Database for U.S. antidumping and safeguard policies.
- U.S. International Trade Commission DataWeb for bilateral imports.
- Broda, Limao, and Weinstein (2008) for foreign export supply elasticities.
- Broda, Greenfield, and Weinstein (2006) for U.S. import demand elasticities.
- U.S. Census Bureau for industry political-economy variables.
- USDA Economic Research Service for bilateral real exchange rates.

Important descriptive facts. Antidumping and safeguard tariff events are rare. In the full sample, the mean antidumping indicator is 0.0017, and the mean antidumping-or-safeguard indicator is 0.0032. Events are more common among top ten trading partners and especially for China.

6 Main Results

6.1 Baseline antidumping results

Table 2 shows that antidumping tariffs are more likely after import growth, more likely in industries with more inelastic demand and supply, and less likely when import growth is historically more volatile. In the baseline probit specification:

- a one-standard-deviation increase in import growth raises the predicted probability of an antidumping tariff from 0.17 percent to 0.23 percent;
- a one-standard-deviation increase in the inverse elasticity measure raises the predicted probability from 0.17 percent to 0.32 percent;
- a one-standard-deviation increase in import-growth volatility lowers the predicted probability from 0.17 percent to 0.04 percent.

These magnitudes correspond to changes of approximately +35 percent, +88 percent, and −76 percent relative to the baseline probability.

6.2 Market-share, safeguards, China, and tariff-size robustness

Table 3 replaces import growth with changes in U.S. market share, adds safeguard tariffs, considers China separately, and estimates the size of antidumping tariffs using a Tobit model. The results are consistent with the theory. In particular, tariffs are more likely when a foreign exporter gains U.S. market share and when the industry elasticity measure implies greater market power.

The China-only results are especially interesting because China accounts for a small share of observations but a large share of U.S. antidumping cases. The estimated effects remain theoretically signed and economically large.

6.3 Political-economy controls

Table 4 adds industry variables often used in political-economy explanations of antidumping and safeguards: four-firm concentration, employment, value-added over shipments, inventories over shipments, and steel/chemical indicators. These variables matter, but the Bagwell-Staiger variables remain important.

Interpretation. The results support a combined view: political economy helps explain which industries are protected, but self-enforcing-agreement logic helps explain when tariff increases become more likely after import shocks and why this varies with market-power variables.

7 Tables from the Paper

7.1 Table 1 and Table 2: Summary Statistics and Baseline Antidumping Tariff Imposition

Read Table 1: tariff events are rare in the full panel, but more common among top trading partners and China. The table also reports the central explanatory variables: import growth, market share changes, the inverse elasticity measure, import-growth volatility, and political-economy controls.

Read Table 2: the baseline antidumping results support the model. Import growth and the inverse elasticity measure increase the probability of tariff imposition, while import-growth volatility lowers it.

Economic message: U.S. antidumping tariffs are not random responses to political pressure alone. They are systematically more likely when the conditions emphasized by self-enforcing trade agreement theory are present.

TABLE 1—SUMMARY STATISTICS: US ANTIDUMPING AND SAFEGUARD TARIFF IMPOSITION

	Full sample		Top 10 trading partners only		China only	
	Mean	SD	Mean	SD	Mean	SD
Dependent variables						
Antidumping (AD) tariff imposed	0.0017	0.0418	0.0046	0.0675	—	—
AD or safeguard tariff imposed	0.0032	0.0562	0.0060	0.0770	0.0323	0.1768
ln(1 + AD tariff)	—	—	0.0030	0.0547	—	—
AD tariff conditional on a positive value	89.7	94.4	116.7	104.0	161.5	99.4
Explanatory variables						
Growth of imports$_{ikt-1}$ [a]	0.102	0.947	0.084	0.567	—	—
Change in US market share$_{ikt-1}$ [b]	0.000	0.004	0.001	0.006	0.005	0.012
$\ln[1/(\eta^f_x + \eta^f_m)]_k$ [c]	−1.991	1.517	−1.995	1.526	−1.982	1.523
$1/(\eta^f_x + \eta^f_m)_k$ [b]	0.241	0.170	—	—	—	—
Standard deviation of import growth$_{ik}$ [b]	0.723	0.660	0.378	0.435	0.393	0.425
Percent change in real exchange rate$_{it-1}$ [c]	0.007	0.116	0.001	0.087	0.017	0.015
Domestic industry variables						
ln(four firm concentration ratio)$_k$ [cd]	3.468	0.608	—	—	—	—
ln(employment)$_{kt-1}$ [cd]	10.377	1.029	—	—	—	—
Value-added/shipments$_{kt-1}$ [cd]	0.513	0.118	—	—	—	—
Inventories/shipments$_{kt-1}$ [cd]	0.129	0.063	—	—	—	—
Indicator for industry k is steel [d]	0.013	0.113	—	—	—	—
Indicator for industry k is chemicals [d]	0.021	0.144	—	—	—	—
Observations	82,341		20,775		2,075	

[a] Rescaled by a factor of 10^{-4} for estimation.
[b] Rescaled by a factor of 10^{-2} for estimation.
[c] Rescaled by a factor of 10^{-3} for estimation.
[d] These variables are based on only 81,943 observations.

(a) Table 1: Summary statistics.

TABLE 2—US ANTIDUMPING TARIFF IMPOSITION:
MARGINAL EFFECTS FROM A BINARY MODEL USING IMPORT GROWTH

	Baseline specification (1)	Substitute alternative elasticity measures (2)	Remove elasticity outliers (3)	Top 10 trading partners only (4)	logit model with multiway clustering (5)
Growth of imports$_{ikt-1}$	4.44***	4.86***	5.66***	28.93***	27.58***
	(1.55)	(1.75)	(1.63)	(8.59)	(9.69)
$\ln[1/(\eta^f_x + \eta^f_m)]_k$	0.58***	—	0.86***	1.36***	1.31*
	(0.14)		(0.20)	(0.39)	(0.75)
$1/(\eta^f_x + \eta^f_m)_k$	—	0.36***	—	—	—
		(0.05)			
Standard deviation of import growth$_{ik}$	−0.16***	−0.18***	−0.18***	−0.54***	−0.54
	(0.02)	(0.02)	(0.03)	(0.16)	(0.45)
Percent change in real exchange rate$_{it-1}$	1.09**	1.15**	1.07*	13.91***	12.05*
	(0.55)	(0.58)	(0.59)	(2.91)	(7.13)
Observations	82,341	82,341	67,262	20,775	20,775
log-likelihood	−1,002.19	−998.17	−857.30	−582.18	−582.23
Predicted probability of antidumping tariff, expressed in percent:[a]					
at means	0.17	0.17	0.19	0.46	0.46
for one standard deviation increase to growth of imports	0.23	0.24	0.27	0.69	0.71
for one standard deviation increase to inverse sum of elasticities	0.32	0.26	0.35	0.73	0.74
for one standard deviation increase to standard deviation of import growth	0.04	0.04	0.04	0.22	0.22
for one standard deviation increase to real exchange rate	0.20	0.20	0.21	0.61	0.60

Notes: Dependent variable is a binary indicator that a US antidumping tariff was imposed on exporting country i in industry k after an investigation initiated in year t. Probit model used to estimate all specifications except for the logit model used to estimate specification (5). Huber-White robust standard errors in parentheses, except for specification (5) which implements Cameron, Gelbach, and Miller (2011) multiway clustering on industry and trading partner.

[a] Predicted probabilities expressed in percent terms; e.g., 0.17 is a predicted probability of seventeen hundredths of one percent, or 0.0017.

*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

(b) Table 2: Baseline antidumping tariff imposition.

7.2 Table 3 and Table 4: Market-Share Robustness and Political-Economy Controls

Read Table 3: replacing import growth with changes in U.S. market share delivers similar results. Including safeguards, focusing on China, and estimating tariff size with a Tobit model also support the model's predictions.

Read Table 4: political-economy controls have the expected signs, but the core variables from Bagwell and Staiger (1990) remain statistically and economically meaningful.

Economic message: the theory survives multiple empirical stress tests. The results are not simply driven by steel, chemicals, or standard industry political-economy variables.

TABLE 3—US ANTIDUMPING AND SAFEGUARD TARIFF IMPOSITION: MARGINAL EFFECTS FROM A BINARY MODEL USING MARKET SHARE

	Substitute change in US market share for import growth (1)	Top 10 trading partners only (2)	AD and safeguard tariff policies[a] (3)	China only[a] (4)	Tobit model with dependent variable as ln(1 + AD tariff) (5)
Change in US market share$_{ikt-1}$	5.48***	14.41***	15.82***	18.28	2,800.09***
	(0.87)	(2.80)	(3.13)	(22.67)	(553.22)
$\ln[1/(\eta^f_x + \eta^f_m)]_k$	0.58***	1.35***	1.86***	6.76**	244.10***
	(0.14)	(0.42)	(0.48)	(2.95)	(77.78)
Standard deviation of import growth$_{ik}$	−0.15***	−0.38***	−0.60***	−3.26***	−73.25***
	(0.02)	(0.12)	(0.15)	(1.06)	(24.64)
Percent change in real exchange rate$_{it-1}$	1.20**	14.82***	22.50***	582.99***	2,777.37***
	(0.61)	(3.08)	(3.56)	(217.19)	(518.14)
Observations	82,341	20,775	20,775	2,075	20,775
log-likelihood	−995.40	−579.51	−716.28	−285.03	−634.95
Predicted probability of antidumping (or safeguard)[a] tariff, expressed in percent:[b]					
at means	0.17	0.46	0.60	3.23	—
for one standard deviation increase to change in US market share	0.20	0.56	0.72	3.44	—
for one standard deviation increase to elasticities	0.32	0.72	0.99	4.38	—
for one standard deviation increase to standard deviation of import growth	0.05	0.27	0.29	1.79	—
for one standard deviation increase to percent change in real exchange rate	0.20	0.61	0.85	4.19	—

Notes: Dependent variable for specifications (1) and (2) is a binary indicator that a US antidumping tariff was imposed on exporting country i in industry k after an investigation initiated in year t. Probit model used to estimate all specifications except for the Tobit model (censored at zero) used to estimate specification (5). Huber-White robust standard errors in parentheses.

[a] Antidumping or safeguard tariff indicator used as dependent variable in specifications (3) and (4).

[b] Predicted probabilities expressed in percent terms; e.g., 0.17 is a predicted probability of seventeen hundredths of one percent, or 0.0017.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

(a) Table 3: Market-share robustness, safeguards, China, and Tobit results.

TABLE 4—US ANTIDUMPING AND SAFEGUARD TARIFF IMPOSITION: IMPORT GROWTH AND INDUSTRY EFFECTS

	Antidumping and safeguard tariff policies	Add political-economy covariates	Add steel and chemical indicators	Predicted probability of antidumping or safeguard tariff for a one standard deviation increase in each explanatory variable, expressed in percent[a]		
	(1)	(2)	(3)	(1)	(2)	(3)
Growth of imports$_{ikt-1}$	6.11***	3.44***	3.34**	0.39	0.38	0.39
	(1.71)	(1.14)	(1.37)			
$\ln[1/(\eta^f_x + \eta^f_m)]_k$	1.19***	0.71***	0.24***	0.66	0.63	0.39
	(0.19)	(0.12)	(0.06)			
Standard deviation of import growth$_{ik}$	−0.25***	−0.14***	−0.16 ***	0.08	0.10	0.09
	(0.03)	(0.02)	(0.02)			
Percent change in real exchange rate$_{it-1}$	4.91***	2.70***	2.75***	0.42	0.40	0.40
	(0.65)	(0.48)	(0.43)			
Domestic industry variables						
ln(four firm conc. ratio)$_k$	—	0.25***	0.13	—	0.38	0.35
		(0.10)	(0.11)			
ln(employment)$_{kt-1}$	—	1.04***	0.70 ***	—	0.62	0.47
		(0.14)	(0.10)			
Value-added/shipments$_{kt-1}$	—	−3.58***	−1.08**	—	0.20	0.28
		(0.64)	(0.54)			
Inventories/shipments$_{kt-1}$	—	6.82***	4.98***	—	0.47	0.41
		(0.98)	(0.75)			
Indicator for industry k is steel	—	—	0.04***	—	—	0.33
			(0.01)			
Indicator for industry k is chemicals	—	—	0.01***	—	—	0.39
			(0.00)			
Predicted probability of antidumping or safeguard tariff, expressed in percent,[a] at means				0.32	0.32	0.32
Observations	81,943	81,943	81,943			
log-likelihood	−1,631.52	−1,512.05	−1,346.50			

Notes: Dependent variable is a binary indicator that a US antidumping tariff or safeguard was imposed on exporting country i in industry k after an investigation initiated in year t. Probit model used to estimate all specifications. Huber-White robust standard errors in parentheses.

[a] Predicted probabilities expressed in percent terms; e.g., 0.32 is a predicted probability of thirty-two hundredths

(b) Table 4: Political-economy controls and industry effects.

8 How to Read the Mechanism

8.1 Why import surges matter

In a repeated game, cooperation survives because defection today is costly in the future. But when imports surge, the immediate terms-of-trade benefit from defecting rises. If the importing country does not receive some tariff flexibility, cooperation may become harder to sustain. Therefore, temporary tariff increases can be interpreted as a way to preserve cooperation rather than destroy it.

8.2 Why elasticities matter

The elasticity term is the central bridge between theory and data. A tariff can improve the terms of trade only when the importing country has market power. If export supply is very elastic or import demand is very elastic, the tariff has little ability to shift prices and mainly creates efficiency losses. Therefore, the probability of a tariff should be larger when

$$\frac{1}{\eta_x^f + \eta_m^f} \tag{9}$$

is larger.

8.3 Why volatility matters

A large import increase does not have the same meaning in every sector. In a sector where imports are usually volatile, a large increase may be normal. In a sector where import growth is typically stable, the same increase is unusual and may create a stronger incentive to defect. The negative coefficient on import-growth volatility is therefore a direct test of the theory.

9 Connection to Broader Research Ideas

9.1 Connection to tariffs as organizational allocation shocks

This paper is directly useful for research ideas about tariffs as shocks that reshape firm and industry incentives. It provides a theory-based way to distinguish tariff increases that occur because of domestic political pressure from tariff increases that arise because the international cooperative constraint becomes harder to sustain. For research designs studying tariffs, this distinction matters because the tariff event is not merely an exogenous policy dummy; it may reveal an underlying state variable: the strength of the temptation to deviate from cooperation.

9.2 Connection to gatekeepers and enforcement institutions

The paper also connects to a broader institutional theme: rules are not self-executing. A trade agreement is a governance arrangement with enforcement constraints. Antidumping and safeguards operate as escape valves within that arrangement. The key economic object is not just the tariff, but the mechanism that lets governments adjust policy without completely collapsing cooperation.

9.3 Connection to empirical IO and policy design

The use of import demand and export supply elasticities is valuable for empirical IO-style work. It shows how structural elasticities can be turned into testable predictions about policy choices. This is useful for studying other policy settings where a government response depends on market power, pass-through, elasticities, or strategic retaliation incentives.

10 Limitations and Open Questions

- **Endogeneity of import growth:** import surges may be correlated with domestic injury or lobbying pressure. The authors add controls, but the design is still observational.
- **Sector-local punishment assumption:** the empirical specification implicitly assumes that retaliation threats are localized by sector. A fully multisector agreement might pool incentive constraints across industries.
- **U.S.-specific setting:** the paper studies the United States from 1997 to 2006. The mechanism may differ for emerging economies or during crisis periods.
- **Temporary trade barriers as cooperative tariffs:** antidumping and safeguards may serve multiple purposes. The interpretation as self-enforcing-agreement adjustment is plausible and empirically supported, but not exclusive.

11 Conclusion

Bown and Crowley provide one of the clearest empirical tests of a repeated-game theory of trade agreements. The central result is that U.S. temporary tariff increases behave in a way consistent with the Bagwell-Staiger model: they are more likely after import surges, more likely when terms-of-trade incentives are stronger, and less likely when import surges are common or volatile.

The paper's deeper message is that trade agreements need not eliminate all tariff variation. Instead, some time-varying protection may be part of the equilibrium structure that keeps cooperation self-enforcing. For researchers studying tariffs, the paper is important because it shows how policy variation can reflect both political forces and international incentive constraints.